

Quadratic Functions & Equations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **identify** the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

RELATED SKILLS

☐ D Determine the vertex and axis of symmetry.☐ D Determine x- and y-intercepts of a quadratic function.☐ A Identify standard, factored, and vertex forms.☐ A Use successive average rates of change to classify concavity.☐ R Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.

☐ 2. I can **connect** standard, factored, and vertex forms to features of a quadratic graph.

RELATED SKILLS

☐ D Connect parameters in a formula to features of its graph.☐ D Identify standard, factored, and vertex forms.☐ A Determine the vertex and axis of symmetry.☐ A Determine x- and y-intercepts of a quadratic function.

☐ 3. I can **rewrite** a quadratic function in a form that reveals a requested feature.

RELATED SKILLS

☐ D Create a perfect-square trinomial by completing the square.☐ D Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.☐ A Factor a monic quadratic trinomial.☐ A Factor a quadratic trinomial with a nonunit leading coefficient.

☐ 4. I can **solve** a quadratic equation by factoring or using the quadratic formula.

RELATED SKILLS

☐ D Select an efficient quadratic-solving method from equation structure.☐ A Apply the zero-product property.☐ A Substitute coefficients accurately into the quadratic formula.☐ A Use the discriminant to predict the number and type of solutions.

☐ **5. I can **graph** a quadratic function using its algebraic structure and key features.**

RELATED SKILLS

- ☐ D Connect parameters in a formula to features of its graph.
 - ☐ D Determine the vertex and axis of symmetry.
 - ☐ D Determine x- and y-intercepts of a quadratic function.
 - ☐ A Identify standard, factored, and vertex forms.
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☐ **6. I can **construct** a quadratic function from specified zeros, a vertex, or graphical information.**

RELATED SKILLS

- ☐ I Build a polynomial from zeros, multiplicities, and a point.
 - ☐ A Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
 - ☐ A Translate quantities and relationships into an equation.
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☐ **7. I can **create and interpret** a quadratic trajectory model.**

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
 - ☐ D Interpret vertex and zeros in an application.
 - ☐ A Attach and interpret units for rates, inputs, and outputs.
 - ☐ A Restrict a model to meaningful contextual inputs.
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